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(54) **SYSTEMS AND METHODS FOR SHARED LATENT SPACE PROMPT TUNING**

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G06F 40/30 (2020.01)
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G06N 3/084 (2023.01)

(52) **U.S. Cl.**
CPC **G06F 40/20** (2020.01); **G06N 3/084** (2013.01)

(58) **Field of Classification Search**

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G06F 40/30; G06N 3/084; G06N 3/044;
G06N 3/045

See application file for complete search history.

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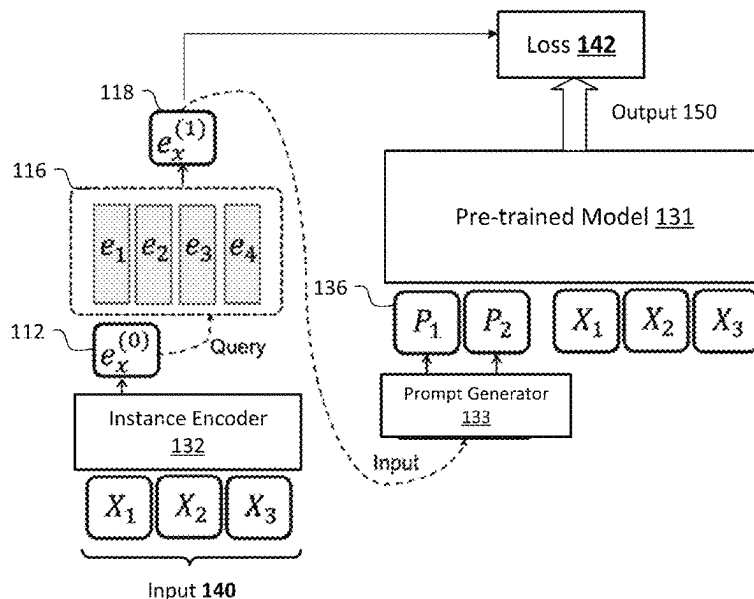
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(57) **ABSTRACT**

Embodiments described herein provide a prompt-based transfer learning method that employs shared latent space prompt tuning). Specifically, a shared latent space is assumed, among all source and target tasks, where each vector in the space captures a basis skill to do a particular task. Given an instance (from either a source task or a target task), it is first encoded into an instance representation vector and then queries the latent space, which yields a skill vector for this instance. This vector modulates a frozen model, via soft prompts which are a simple prompt transformation (the prompt generator in FIG. 3) of the basis skill vector, to generate an answer for the instance. The latent space and prompt transformation are learned end-to-end in upstream pre-training on source tasks.

16 Claims, 7 Drawing Sheets



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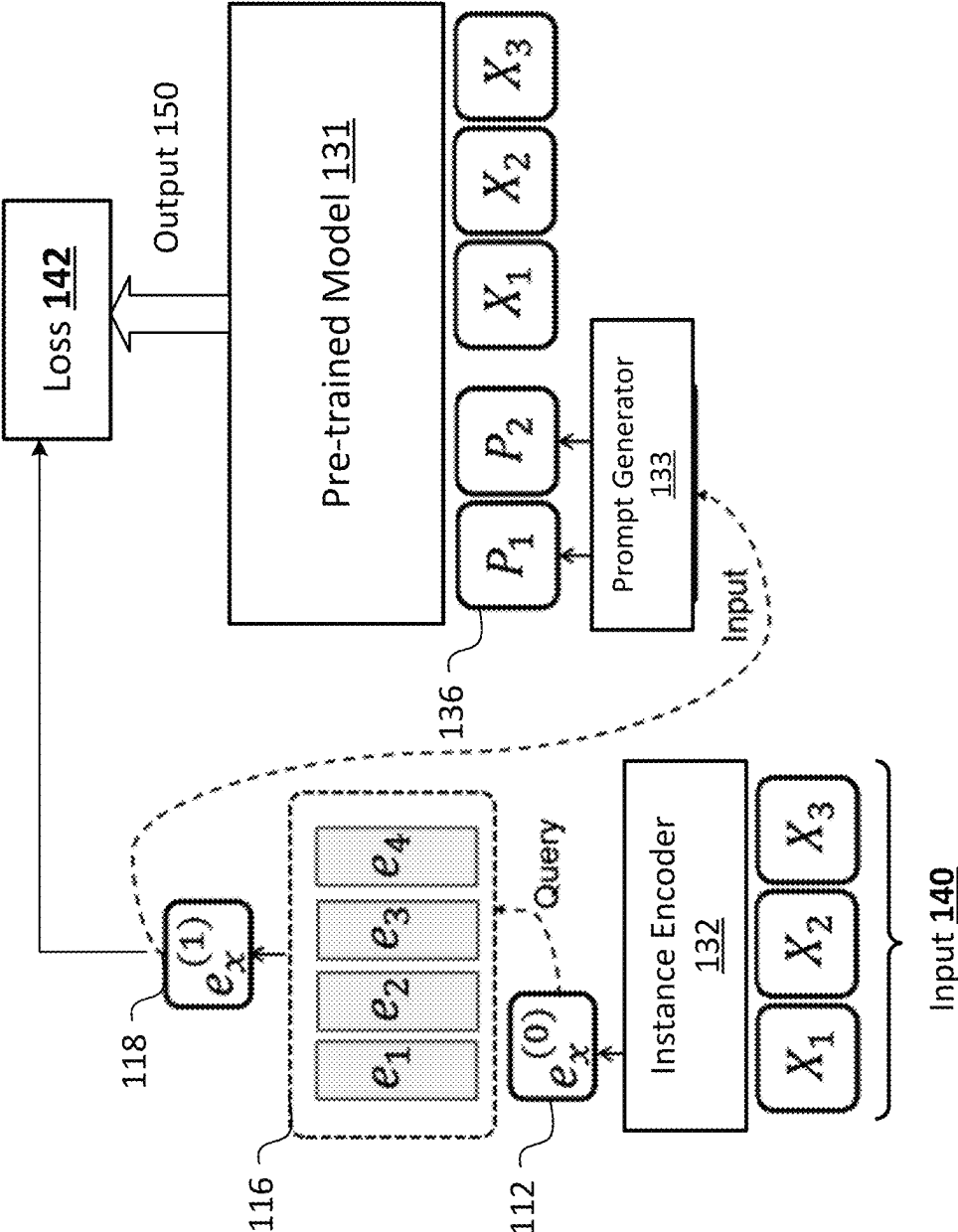


FIG. 1

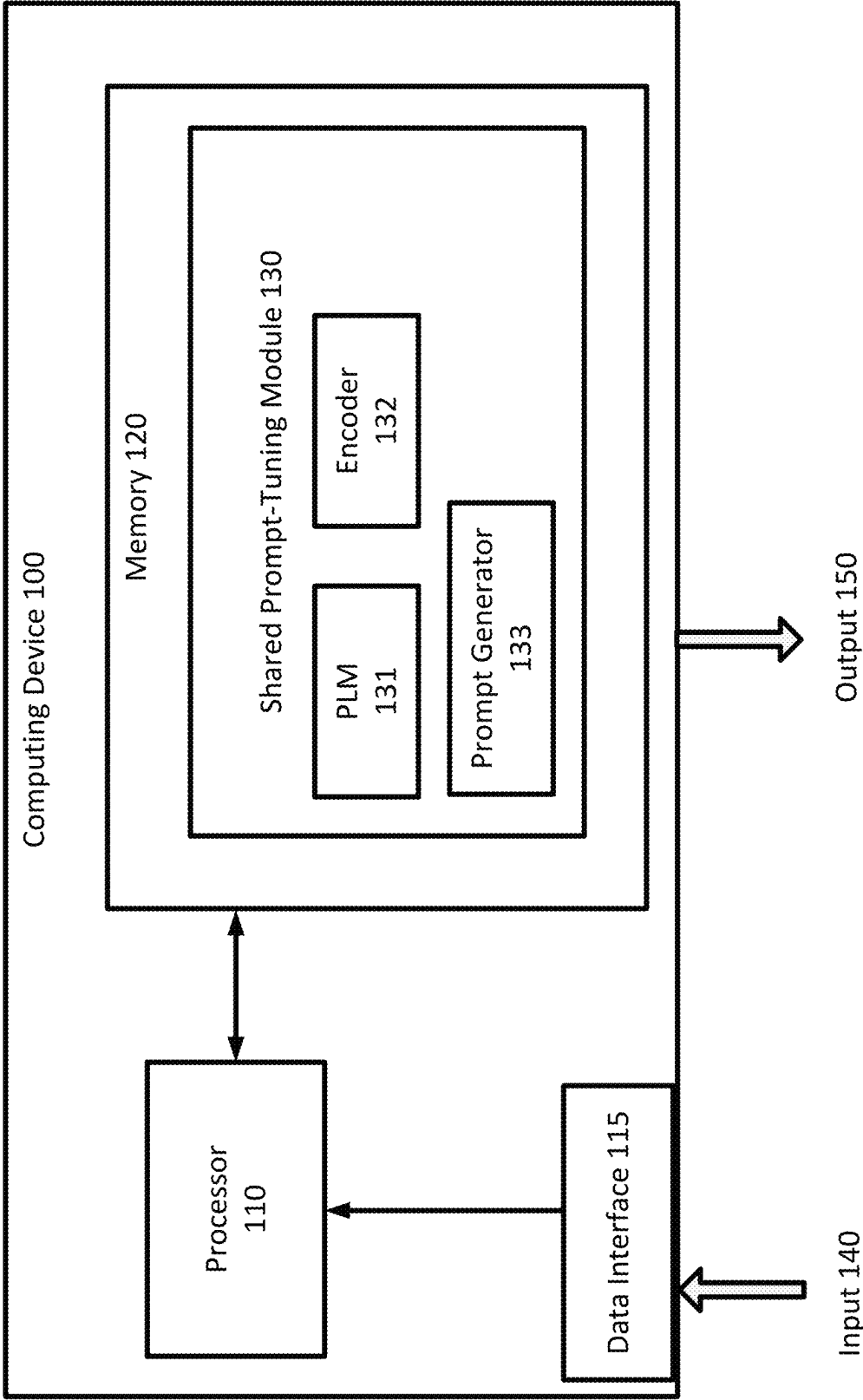


FIG. 2

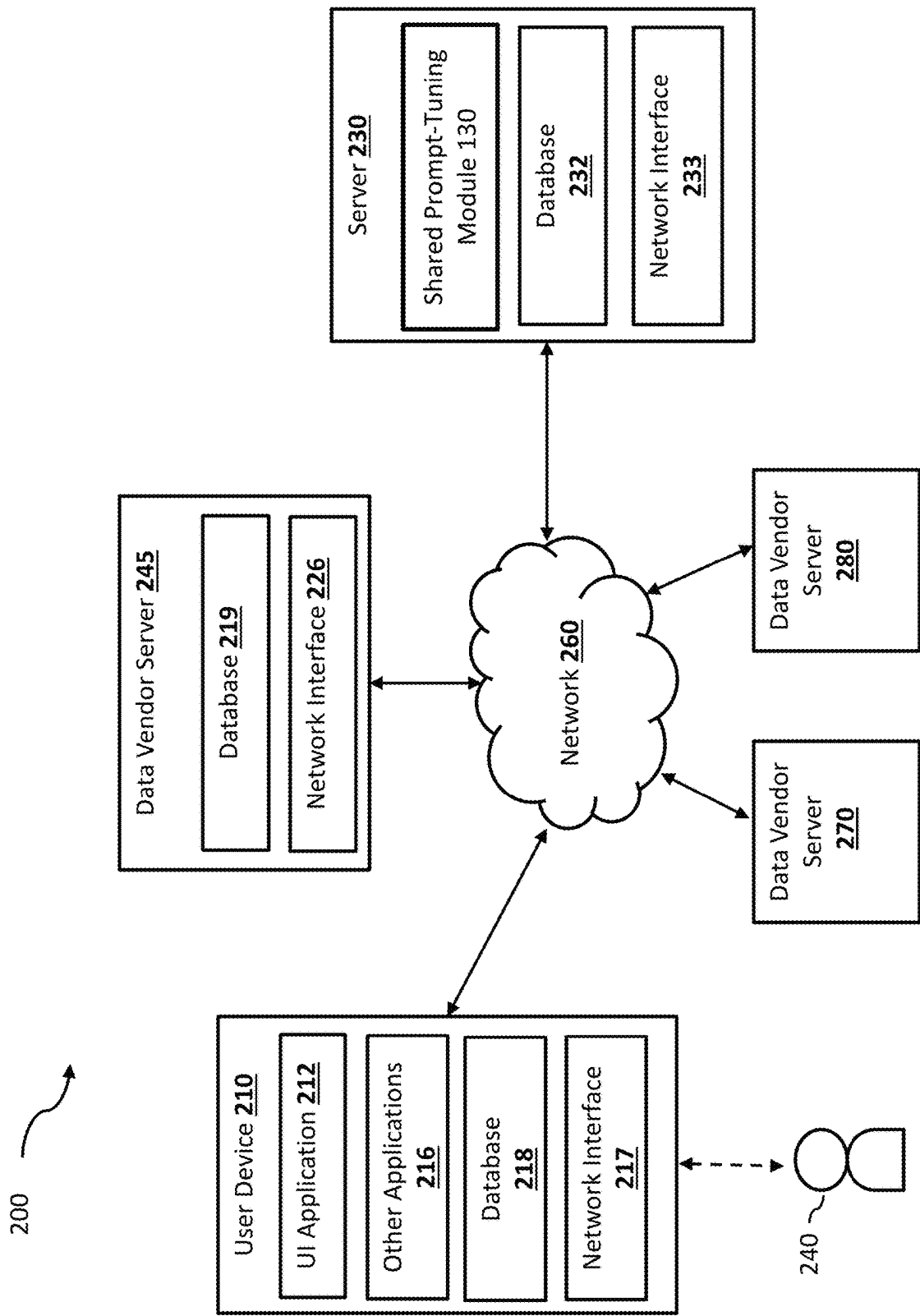
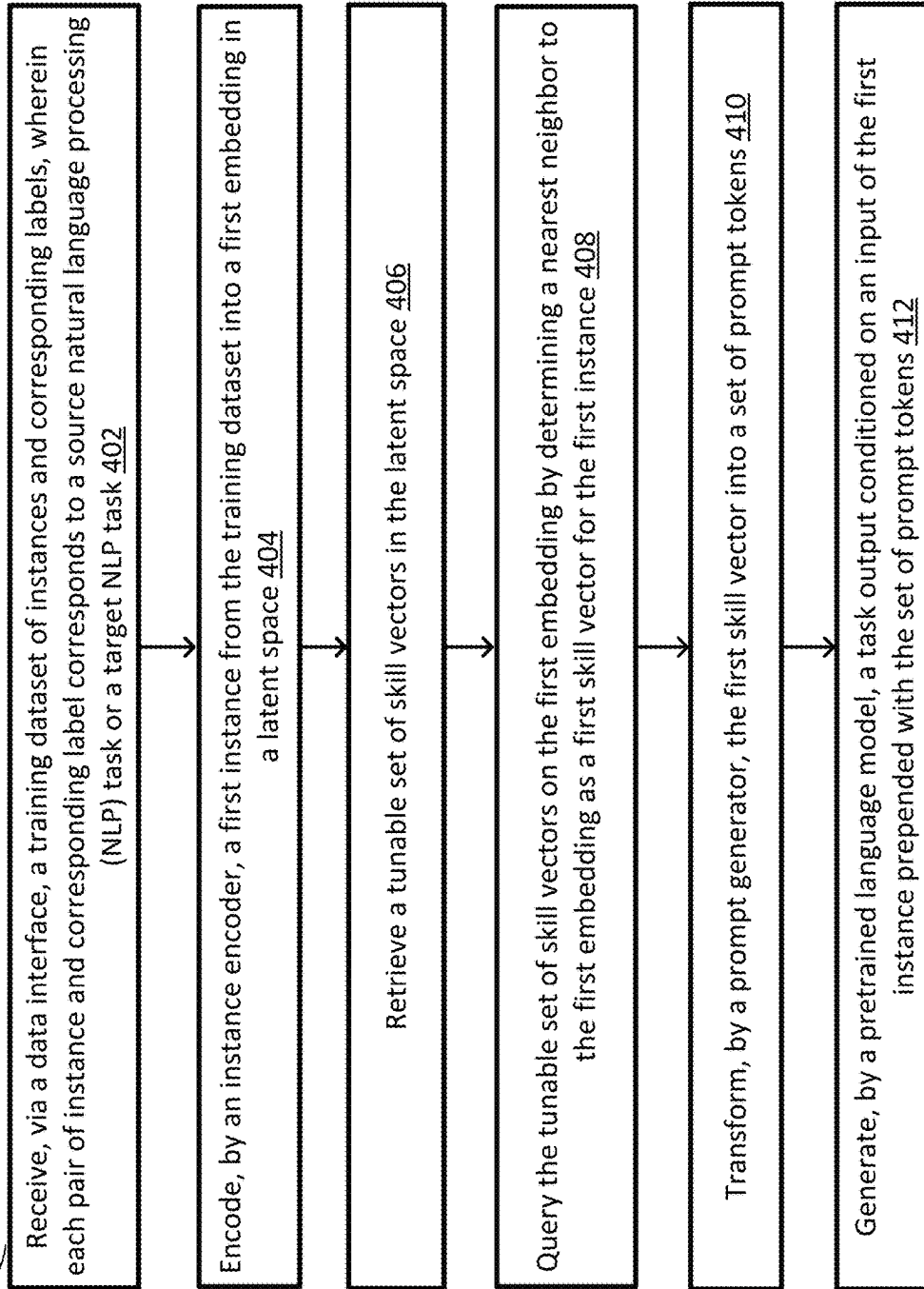


FIG. 3

400



414

FIG. 4A

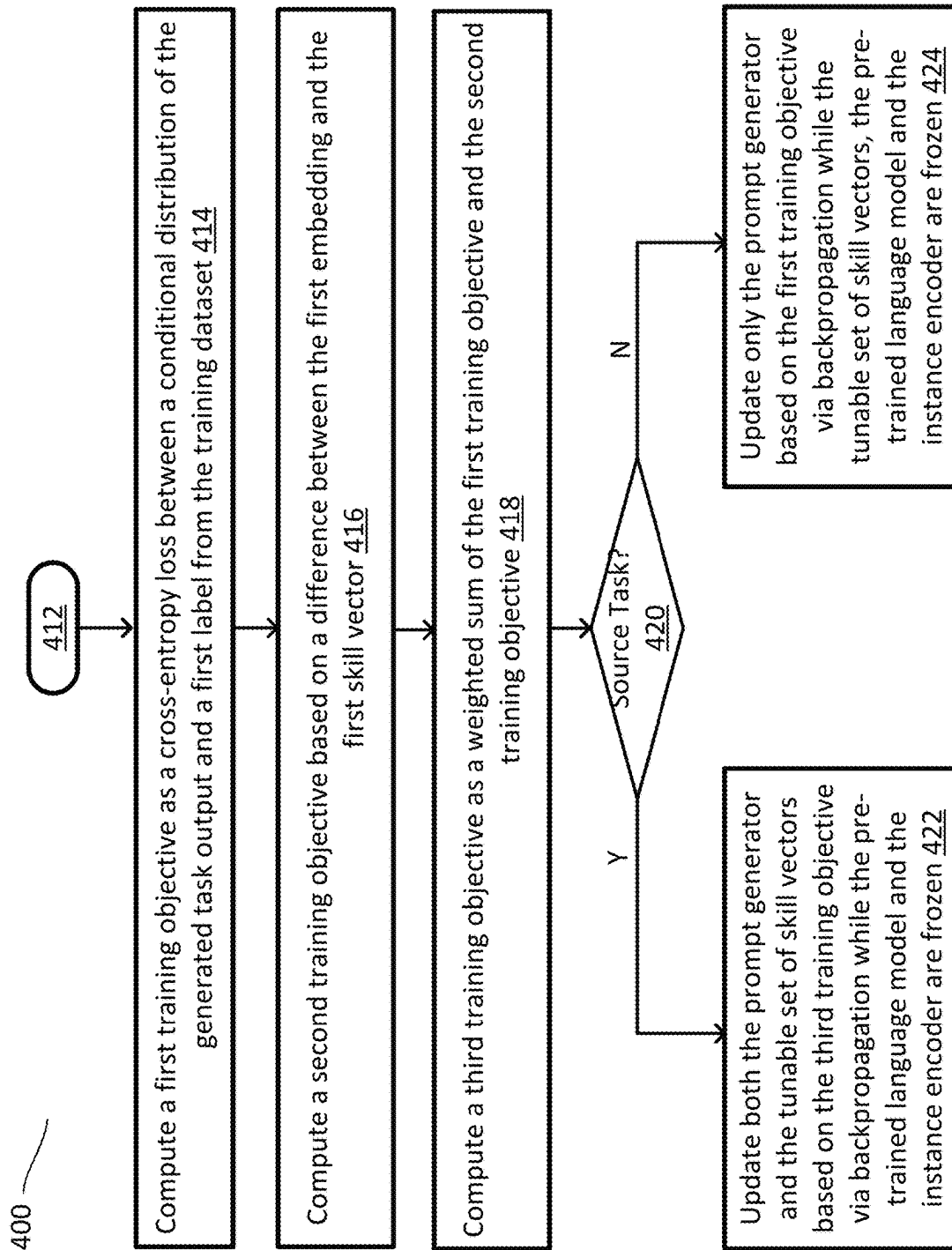


FIG. 4B

	BoolQ	CB	CoLA	COPA	CR	MRPC	RTE	STS-B	WiC	WSC	Average
ModelTuning	81.4	94.0	51.1	71.2	<u>94.1</u>	87.5	81.5	<u>89.4</u>	68.3	<u>80.8</u>	<u>79.9</u>
SPoT-Oracle	77.6	97.0	<u>55.6</u>	<u>69.3</u>	93.9	<u>88.7</u>	74.7	90.0	70.2	77.2	79.4
PromptTuning	73.0	92.7	52.9	56.7	93.5	86.1	68.7	88.1	63.6	71.5	74.7
SPoT-Retrieval	74.2	95.4	54.8	58.3	93.6	88.4	71.6	90.0	66.7	72.9	76.6
SHARPT	<u>78.9</u>	<u>94.6</u>	58.2	67.0	94.5	89.7	<u>79.4</u>	89.1	<u>68.8</u>	81.6	80.2

Table 1: Results on the high-to-low transfer learning setting. Methods in the upper panel are significantly more expensive than those in the lower panel. The best performance is in **bold**, and the second best is underlined.

FIG. 5

	COPA	CB	RTE	WSC	WiC
ModelTuning	71.2	<u>94.0</u>	81.5	80.8	68.3
SPoT-Oracle	63.0	92.9	72.0	77.2	70.2
PromptTuning	56.7	92.7	68.7	71.5	63.6
SPoT-Retrieval	61.2	89.4	71.4	73.6	66.7
SHARPT	<u>65.0</u>	94.6	<u>79.4</u>	<u>79.0</u>	<u>69.8</u>

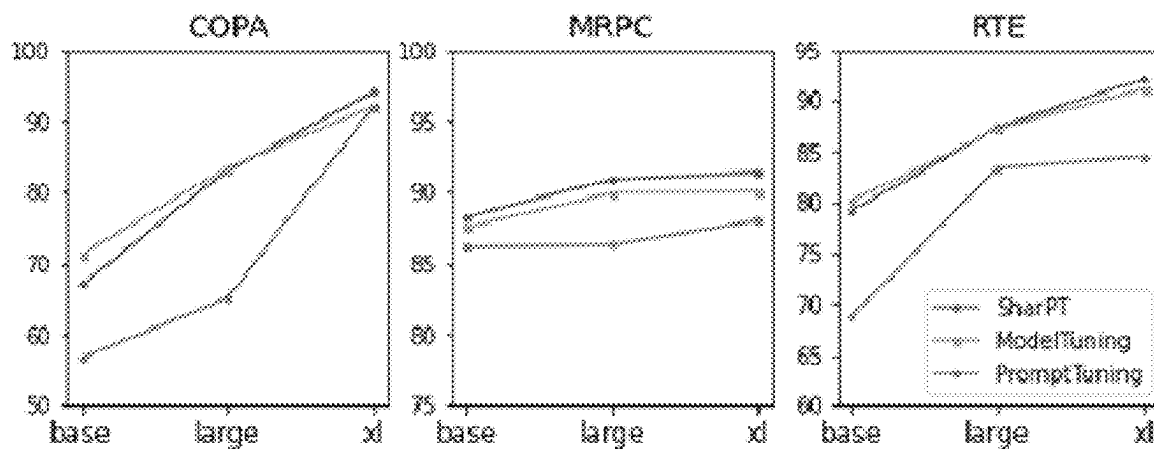
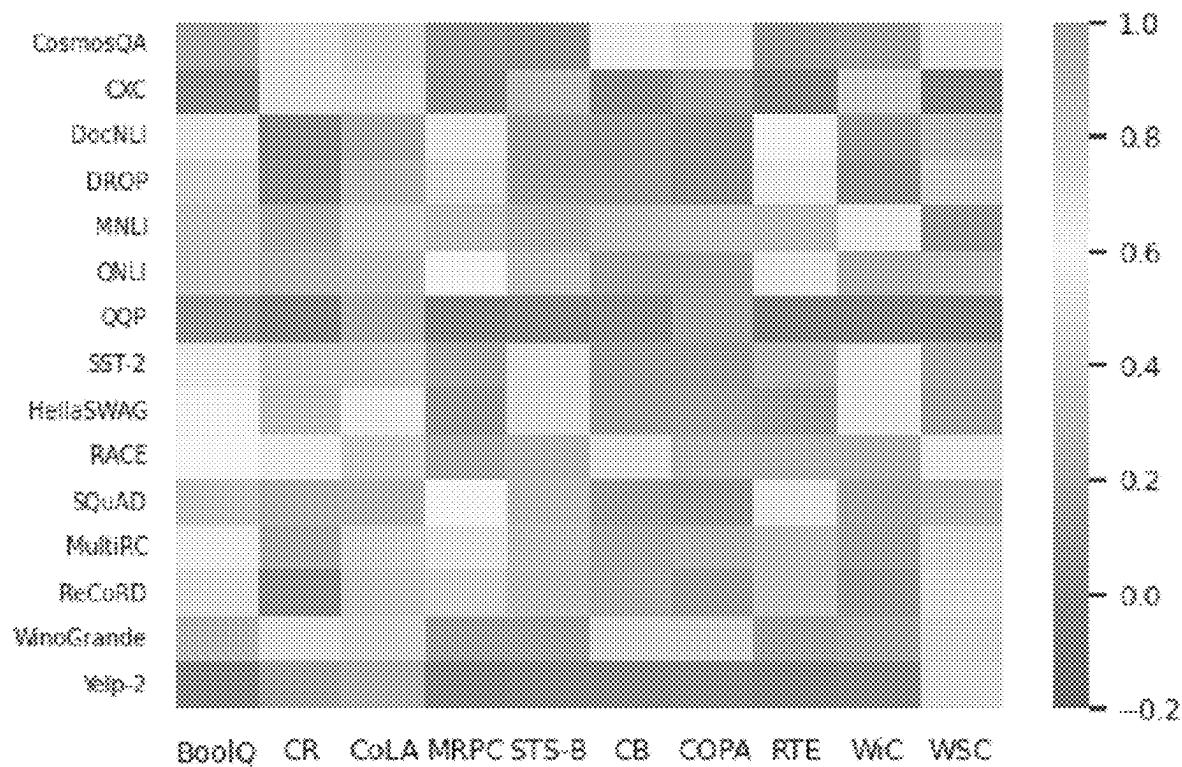
Table 2: Results on transferring across task categories.

FIG. 6

	BoolQ	CB	CoLA	COPA
SHARPT	78.9	94.6	58.2	67.0
No Source Task Training	64.3	89.3	10.3	58.0
No Latent Space	67.9	82.4	17.6	61.0

Table 3: Ablation results.

FIG. 7

**FIG. 8****FIG. 9**

SYSTEMS AND METHODS FOR SHARED LATENT SPACE PROMPT TUNING

CROSS REFERENCES

The application is a nonprovisional of and claims priority to U.S. Provisional application No. 63/355,468, filed Jun. 24, 2022, which is hereby expressly incorporated by reference herein in its entirety.

TECHNICAL FIELD

The embodiments relate generally to natural language processing (NLP) models and machine learning systems, and more specifically to systems and methods for shared latent space prompt tuning in language models.

BACKGROUND

Pre-trained language models (PLMs) may be finetuned to perform natural language processing tasks, such as question answering, sentiment classification, summarization, and/or the like. The finetuning often entails training the PLMs based on a training dataset that is specific to a certain NLP task, including updating all parameter weights of a PLM for a downstream task (i.e., MODEL TUNING). This approach is however prohibitively expensive, especially given the growing sizes of PLMs, which may include tens of thousands of parameters.

Therefore, there is a need for efficient methods to adapt PLMs in various natural language processing tasks.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an example block diagram illustrating a shared latent space prompt tuning framework, according to embodiments described herein.

FIG. 2 is a simplified diagram illustrating a computing device implementing the shared prompt-tuning model described in FIG. 1, according to one embodiment described herein.

FIG. 3 is a simplified diagram illustrating a networked system implementing the shared prompt-tuning model described in FIG. 1, according to one embodiment described herein.

FIGS. 4A-4B provide an example logic flow diagram illustrating a method of shared prompt-tuning in language models, according to some embodiments described herein.

FIGS. 5-9 provide example data charts and performance plots illustrating example performance of the shared prompt-tuning framework described herein.

Embodiments of the disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures, wherein showings therein are for purposes of illustrating embodiments of the disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

As used herein, the term “network” may comprise any hardware or software-based framework that includes any artificial intelligence network or system, neural network or system and/or any training or learning models implemented thereon or therewith.

As used herein, the term “module” may comprise hardware or software-based framework that performs one or more functions. In some embodiments, the module may be implemented on one or more neural networks.

Finetuning all parameter weights of a PLM on a downstream task (i.e., MODEL TUNING) can be prohibitively expensive, especially given the growing sizes of PLMs. Instead of finetuning all parameters of language models for each downstream task, prompt models may be used to modulate a frozen LLM for a downstream task. For example, the input training text may be prepended with a prompt, which often consists of a task description and several examples, e.g., “the user intends to {check account balance, make a deposit, dispute a charge}.” Such prompts can effectively adapt a large PLM, such as a frozen GPT-3. This approach, however, does not achieve comparable performance with MODEL TUNING and is sensitive to choice of the wording of the prompt. Alternatively, instead of actual text (or hard prompt), a soft prompt, consisting of k tunable tokens, may be prepended to the input text. The soft prompt can be optimized with gradient-based methods for each downstream task. Soft prompt tuning may achieve comparable performance with MODEL TUNING when the model size is large (e.g., over 10B parameters) but still underperforms with smaller models.

Some other existing approaches may leverage knowledge as encoded in soft prompts, shared across tasks. In particular, a task-specific soft prompt may be learnt for each task in a set of source tasks. Given a target task, the set of source prompts are searched and the best one or some weighted combination of the searches are used to initialize the prompt for the target task and then to tune the prompt. While these methods may further improve the performance with smaller models (e.g., 200M parameters), tuning and search prompts can be complicated and expensive to identify the source task that provides optimal prompts.

Embodiments described herein provide a prompt-based transfer learning method that employs shared latent space prompt tuning. Specifically, a shared latent space is assumed, among all source and target tasks, where each vector in the space captures a basis skill to do a particular task. Given an instance (from either a source task or a target task), it is first encoded into an instance representation vector and then queries the latent space, which yields a skill vector for this instance. This vector modulates a frozen model, via soft prompts which are a simple prompt transformation (the prompt generator in FIG. 3) of the basis skill vector, to generate an answer for the instance. The latent space and prompt transformation are learned end-to-end in upstream pre-training on source tasks.

In this way, the shared prompt-tuning improves parameter-efficiency by only updating a small amount of, e.g., 0.1% to 0.3% parameters of the frozen backbone model. As the instance encoder is always frozen, the embeddings of all instances can be pre-computed and only keep the embeddings. Thus, memory and time can be saved.

Overview

FIG. 1 is an example block diagram illustrating a shared latent space prompt tuning framework, according to embodiments described herein. The prompt tuning framework shows an instance encoder 132, a prompt generator 133 and a pre-trained language model (PLM) 131 that are communicatively coupled and may be operated concurrently, independently or inter-dependently.

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Traditionally, given a NLP task (e.g., summarization, question answering, intent classification, etc.) and corresponding training dataset $T=\{(x,y)\}$ and a PLM such as **131**, the traditional MODULE TUNING method updates all parameters θ of the PLM to minimize a loss objective: $L(\theta)=-\log \log P_\theta(x)$, e.g., to maximize the conditional probability of generating an output y given an input x .

In the shared latent space prompt tuning framework as shown in FIG. 1, an input instance x **140** from the training set, as denoted by $\{X1, X2, X3\}$ (e.g., a sequence of NLP tokens), is encoded by the instance encoder **132** into an encoded vector **112**, denoted by $e_x^{(0)}$. The discrete latent space **116** of the encoded vector $e_x^{(0)}$ consists of a set of skill vectors $E=\{e_i \in R^m\}_{i=1}^K$. Given the input instance x , the encoded vector **112** may then be used to query the latent space E to find the nearest neighbor to the encoded vector $e_x^{(0)}$, resulting in a skill vector $e_x^{(1)}$ **118**:

$$e_x^{(1)} = e_k, k = \arg \min_{i \in [K]} \|e_x^{(0)} - e_i\|_2.$$

The skill vector $e_x^{(1)}$ **118** is then transformed by a light-weight prompt generator **133** to generate soft prompt $p=f_\alpha(e_x^{(1)})$. For example, the prompt generator f_α is a light-weight multi-layer perceptron (MLP) that transforms the skill vector by:

$$e_x^{(1)} = \text{Tan } h(W_1 e_x^{(1)} + b_1), p_l = W_2(z_l + e_x^{(1)}) + b_2,$$

where $z_l \in R^d$ is the position embedding for the l th token and $W_1 \in R^{d \times m}$, $W_2 \in R^{d \times d}$. Thus, the soft prompt generator **133** generates $p=\{p_l\}_{l=1}^L$ soft prompts, where $p \in R^{L \times d}$ has L vectors of size d .

The prompt tokens **136** are represented by $\{P1, P2\}$ shown in FIG. 1. It is noted that soft tokens $\{P1, P2\}$ are for illustrative purpose only. Any different number of tokens may be used for the soft prompts. The prompt tokens **136** are prepended to the instance tokens **140** to form an input sequence to the pre-trained model **131**. The pre-trained model **131** then generates a task-specific output y **150**. Then, for a target task, the loss module **142** computes a loss:

$$L(\alpha)=-\log \log P_\theta(f_\alpha(e_k), x).$$

In target task training aforementioned, the latent space E **116** is known and fixed. However, for source tasks, the latent space needs to be learnt. For example, if the training dataset comprises data corresponding to N sources tasks, $\{T_j^{(s)}\}_{j=1}^N$, the set of data of all tasks may be mixed together, $T^{(s)}=\bigcup_{j=1}^N T_j^{(s)}$. Given $x \in T^{(s)}$, its embedding $e_x^{(0)}$ may be generated by the instance encoder **132**. Then, without knowing the latent space for $e_x^{(0)}$, a loss is computed to learn the latent space E :

$$L(E)=\|sg(e_x^{(0)})-e_k\|_2,$$

where $sg(\cdot)$ is a stop gradient operator and e_k is the nearest neighbor for $e_x^{(0)}$ as discussed above. The overall loss in source task learning is,

$$L(\alpha, E)=L(\alpha)+L(E).$$

Therefore, the forward pass for training on source and target tasks are exactly the same. For source task training data, the training loss objective is computed as $L(\alpha, E)$, while for target task training data, the loss is computed as $L(\alpha)$.

After the loss module **142** computes the respective losses, the instance encoder **132** and the pre-trained model **131** are frozen in all scenarios. The skill vectors in the latent space

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116 and the prompt generator **133** are updated in source task training based on $L(\alpha, E)$. The skill vectors in the latent space **116** is frozen in target task training (as defined for each individual target task), and the prompt generator is updated based on $L(\alpha)$.

In this way, the shared latent space prompt tuning framework may train the prompt generator for a diverse set of tasks covering diverse domains and task categories. Specifically, a shared latent skill space which is learned end-to-end from a mixture of all source tasks in a single training run. Also, the tuning on the target task only requires a single run.

Computer and Networked Environments

FIG. 2 is a simplified diagram illustrating a computing device implementing the shared prompt-tuning model described in FIG. 1, according to one embodiment described herein. As shown in FIG. 2, computing device **100** includes a processor **110** coupled to memory **120**. Operation of computing device **100** is controlled by processor **110**. And although computing device **100** is shown with only one processor **110**, it is understood that processor **110** may be representative of one or more central processing units, multi-core processors, microprocessors, microcontrollers, digital signal processors, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), graphics processing units (GPUs) and/or the like in computing device **100**. Computing device **100** may be implemented as a stand-alone subsystem, as a board added to a computing device, and/or as a virtual machine.

Memory **120** may be used to store software executed by computing device **100** and/or one or more data structures used during operation of computing device **100**. Memory **120** may include one or more types of machine-readable media. Some common forms of machine-readable media may include floppy disk, flexible disk, hard disk, magnetic tape, any other magnetic medium, CD-ROM, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, RAM, PROM, EPROM, FLASH-EPROM, any other memory chip or cartridge, and/or any other medium from which a processor or computer is adapted to read.

Processor **110** and/or memory **120** may be arranged in any suitable physical arrangement. In some embodiments, processor **110** and/or memory **120** may be implemented on a same board, in a same package (e.g., system-in-package), on a same chip (e.g., system-on-chip), and/or the like. In some embodiments, processor **110** and/or memory **120** may include distributed, virtualized, and/or containerized computing resources. Consistent with such embodiments, processor **110** and/or memory **120** may be located in one or more data centers and/or cloud computing facilities.

In some examples, memory **120** may include non-transitory, tangible, machine readable media that includes executable code that when run by one or more processors (e.g., processor **110**) may cause the one or more processors to perform the methods described in further detail herein. For example, as shown, memory **120** includes instructions for shared prompt-tuning module **130** that may be used to implement and/or emulate the systems and models, and/or to implement any of the methods described further herein. A shared prompt-tuning module **130** may receive input **140** such as a text input via the data interface **115** and generate an output **150** which may be an answer to the question, and/or the like. Examples of the input data may include a

textual question, a sentence and/or the like. Examples of the output data may include a text classification, an answer, and/or the like.

The data interface **115** may comprise a communication interface, a user interface (such as a voice input interface, a graphical user interface, and/or the like). For example, the computing device **100** may receive the input **140** (such as a training dataset) from a networked database via a communication interface. Or the computing device **100** may receive the input **140**, such as an articulated question, from a user via the user interface.

In some embodiments, the shared prompt-tuning module **130** is configured to be trained for various NLU tasks. The shared prompt-tuning module **130** may further include a pretrained PLM submodule **131**, an instance encoder **132** and a prompt generator submodule **133**, which are all further described in relation to FIG. **3**. In one embodiment, the shared prompt-tuning module **130** and its submodules **131-133** may be implemented by hardware, software and/or a combination thereof.

Some examples of computing devices, such as computing device **100** may include non-transitory, tangible, machine readable media that include executable code that when run by one or more processors (e.g., processor **110**) may cause the one or more processors to perform the processes of method. Some common forms of machine-readable media that may include the processes of method are, for example, floppy disk, flexible disk, hard disk, magnetic tape, any other magnetic medium, CD-ROM, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, RAM, PROM, EPROM, FLASH-EPROM, any other memory chip or cartridge, and/or any other medium from which a processor or computer is adapted to read.

FIG. **3** is a simplified diagram illustrating a networked system implementing the shared prompt-tuning framework described in FIG. **1**, according to one embodiment described herein. In one embodiment, block diagram **200** shows a system including the user device **210** which may be operated by user **240**, data vendor servers **245**, **270** and **280**, server **230**, and other forms of devices, servers, and/or software components that operate to perform various methodologies in accordance with the described embodiments. Exemplary devices and servers may include device, stand-alone, and enterprise-class servers which may be similar to the computing device **100** described in FIG. **1**, operating an OS such as a MICROSOFT® OS, a UNIX® OS, a LINUX® OS, or other suitable device and/or server-based OS. It can be appreciated that the devices and/or servers illustrated in FIG. **2** may be deployed in other ways and that the operations performed, and/or the services provided by such devices and/or servers may be combined or separated for a given embodiment and may be performed by a greater number or fewer number of devices and/or servers. One or more devices and/or servers may be operated and/or maintained by the same or different entities.

The user device **210**, data vendor servers **245**, **270** and **280**, and the server **230** may communicate with each other over a network **260**. User device **210** may be utilized by a user **240** (e.g., a driver, a system admin, etc.) to access the various features available for user device **210**, which may include processes and/or applications associated with the server **230** to receive an output data anomaly report.

User device **210**, data vendor server **245**, and the server **230** may each include one or more processors, memories, and other appropriate components for executing instructions such as program code and/or data stored on one or more

computer readable mediums to implement the various applications, data, and steps described herein. For example, such instructions may be stored in one or more computer readable media such as memories or data storage devices internal and/or external to various components of system **200**, and/or accessible over network **260**.

User device **210** may be implemented as a communication device that may utilize appropriate hardware and software configured for wired and/or wireless communication with data vendor server **245** and/or the server **230**. For example, in one embodiment, user device **210** may be implemented as an autonomous driving vehicle, a personal computer (PC), a smart phone, laptop/tablet computer, wristwatch with appropriate computer hardware resources, eyeglasses with appropriate computer hardware (e.g., GOOGLE GLASS®), other type of wearable computing device, implantable communication devices, and/or other types of computing devices capable of transmitting and/or receiving data, such as an IPAD® from APPLE®. Although only one communication device is shown, a plurality of communication devices may function similarly.

User device **210** of FIG. **2** contains a user interface (UI) application **212**, and/or other applications **216**, which may correspond to executable processes, procedures, and/or applications with associated hardware. For example, the user device **210** may receive a message indicating an answer to a question from the server **230** and display the message via the UI application **212**. In other embodiments, user device **210** may include additional or different modules having specialized hardware and/or software as required.

In various embodiments, user device **210** includes other applications **216** as may be desired in particular embodiments to provide features to user device **210**. For example, other applications **216** may include security applications for implementing client-side security features, programmatic client applications for interfacing with appropriate application programming interfaces (APIs) over network **260**, or other types of applications. Other applications **216** may also include communication applications, such as email, texting, voice, social networking, and IM applications that allow a user to send and receive emails, calls, texts, and other notifications through network **260**. For example, the other application **216** may be an email or instant messaging application that receives a result message from the server **230**. Other applications **216** may include device interfaces and other display modules that may receive input and/or output information. For example, other applications **216** may contain software programs for asset management, executable by a processor, including a graphical user interface (GUI) configured to provide an interface to the user **240** to view the answer.

User device **210** may further include database **218** stored in a transitory and/or non-transitory memory of user device **210**, which may store various applications and data and be utilized during execution of various modules of user device **210**. Database **218** may store user profile relating to the user **240**, predictions previously viewed or saved by the user **240**, historical data received from the server **230**, and/or the like. In some embodiments, database **218** may be local to user device **210**. However, in other embodiments, database **218** may be external to user device **210** and accessible by user device **210**, including cloud storage systems and/or databases that are accessible over network **260**.

User device **210** includes at least one network interface component **219** adapted to communicate with data vendor server **245** and/or the server **230**. In various embodiments, network interface component **219** may include a DSL (e.g.,

Digital Subscriber Line) modem, a PSTN (Public Switched Telephone Network) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency, infrared, Bluetooth, and near field communication devices.

Data vendor server **245** may correspond to a server that hosts one or more of the databases **203a-n** (or collectively referred to as **203**) to provide training datasets including training images and questions to the server **230**. The database **203** may be implemented by one or more relational database, distributed databases, cloud databases, and/or the like.

The data vendor server **245** includes at least one network interface component **226** adapted to communicate with user device **210** and/or the server **230**. In various embodiments, network interface component **226** may include a DSL (e.g., Digital Subscriber Line) modem, a PSTN (Public Switched Telephone Network) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency, infrared, Bluetooth, and near field communication devices. For example, in one implementation, the data vendor server **245** may send asset information from the database **203**, via the network interface **226**, to the server **230**.

The server **230** may be housed with the shared prompt-tuning module **130** and its submodules described in FIG. 2. In some implementations, module **130** may receive data from database **219** at the data vendor server **245** via the network **260** to generate an answer to a visual question. The generated answer may also be sent to the user device **210** for review by the user **240** via the network **260**.

The database **232** may be stored in a transitory and/or non-transitory memory of the server **230**. In one implementation, the database **232** may store data obtained from the data vendor server **245**. In one implementation, the database **232** may store parameters of the multilingual model **130**. In one implementation, the database **232** may store previously generated answers, and the corresponding input feature vectors.

In some embodiments, database **232** may be local to the server **230**. However, in other embodiments, database **232** may be external to the server **230** and accessible by the server **230**, including cloud storage systems and/or databases that are accessible over network **260**.

The server **230** includes at least one network interface component **233** adapted to communicate with user device **210** and/or data vendor servers **245**, **270** or **280** over network **260**. In various embodiments, network interface component **233** may comprise a DSL (e.g., Digital Subscriber Line) modem, a PSTN (Public Switched Telephone Network) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency (RF), and infrared (IR) communication devices.

Network **260** may be implemented as a single network or a combination of multiple networks. For example, in various embodiments, network **260** may include the Internet or one or more intranets, landline networks, wireless networks, and/or other appropriate types of networks. Thus, network **260** may correspond to small scale communication networks, such as a private or local area network, or a larger

scale network, such as a wide area network or the Internet, accessible by the various components of system **200**.

Example Work Flows

FIGS. 4A-4B provide an example logic flow diagram illustrating a method of shared prompt-tuning in language models, according to some embodiments described herein. One or more of the processes of method **400** may be implemented, at least in part, in the form of executable code stored on non-transitory, tangible, machine-readable media that when run by one or more processors may cause the one or more processors to perform one or more of the processes. In some embodiments, method **400** corresponds to the operation of the shared prompt-tuning module **230** (e.g., FIGS. 2-3).

At step **402**, a training dataset of instances and corresponding labels are received, via a data interface (e.g., **215** in FIG. 2). Each pair of instance and corresponding label corresponds to a source natural language processing (NLP) task or a target NLP task.

At step **404**, an instance encoder (e.g., **132** in FIG. 1) may encode a first instance from the training dataset into a first embedding (e.g., **112** in FIG. 1) in a latent space (e.g., **116** in FIG. 1).

At step **406**, a tunable set of skill vectors are retrieved in the latent space (e.g., **116** in FIG. 1).

At step **408**, the first embedding (e.g., **112** in FIG. 1) may query the tunable set of skill vectors on the first embedding by determining a nearest neighbor to the first embedding as a first skill vector (e.g., **118** in FIG. 1) for the first instance.

At step **410**, a prompt generator (e.g., **133** in FIG. 1) may transform the first skill vector into a set of prompt tokens (e.g., **136** in FIG. 1).

At step **412**, a pretrained language model (e.g., **131** in FIG. 1) may generate a task output (e.g., **150** in FIG. 1) conditioned on an input of the first instance (e.g., **140** in FIG. 1) prepended with the set of prompt tokens (e.g., **136** in FIG. 1).

At step **414**, a first training objective (e.g., $L(a)$) may be computed as a cross-entropy loss between a conditional distribution of the generated task output and a first label from the training dataset.

At step **416**, a second training objective (e.g., $L(E)$) may be computed based on a difference between the first embedding and the first skill vector.

At step **418**, a third training objective (e.g., $L(\alpha, E)$) may be computed as a weighted sum of the first training objective and the second training objectives.

At step **420**, the method **400** determines whether the first instance belongs to a source task. When the first instance is selected from a mixture of source NLP tasks, method **400** proceeds to step **422**, where both the prompt generator and the tunable set of skill vectors are updated based on the third training objective (e.g., $L(\alpha, E)$) via backpropagation while the pre-trained language model and the instance encoder are frozen. When the first instance is selected only for the target NLP task, method **400** proceed to step **424**, where only the prompt generator is updated based on the first training objective (e.g., $L(\alpha)$) via backpropagation while the tunable set of skill vectors, the pre-trained language model and the instance encoder are frozen.

Example Data Experiments Performance

For high-to-low resource transfer, the target tasks may be low-resource tasks (less than 10K training examples), while

the source tasks are high-resource tasks. The testing dataset consists of 25 tasks in total. There are 15 source tasks (e.g., DocNLI, DROP) and 10 target tasks (e.g., BoolQ, CoLA). The data test setting are kept almost the same as a major experiment in Vu et al., SPoT: Better frozen model adaptation through soft prompt transfer, in Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 5039-5059, 2022, for a fair comparison, with the exception that C4 is excluded from the source task since it is much a larger dataset than other tasks. Excluding C4 does not affect SPOT performance since it does not provide an optimal source prompt for any target task.

In transfer across different task categories, the transferability from datasets is investigated in some task categories to datasets in other held-out task categories. Following Sanh et al., Multi-task prompted training enables zero-shot task generalization, in International Conference on Learning Representations, 2022, datasets are assumed in each category to measure a general NLP ability, and use the same taxonomy defined in Sanh et al. The source tasks include (1) QA tasks: ReCoRD, SQuAD, DROP, MultiRC, and RACE; (2) sentiment analysis tasks: Yelp-2 and SST-2; (3) a paraphrase detection task: QQP; (4) a semantic similarity task: CXC. The target tasks include (1) a sentence completion task: COPA; (2) NLI tasks: CB and RTE; (3) a coreference resolution tasks: WSC; (4) a word sense disambiguation task: WiC.

In training, all datasets are converted to a text-to-text format. All experiments are conducted with T5-base as the backbone unless stated.

SimCSE (Gao et al., SimCSE: Simple contrastive learning of sentence embeddings, in Empirical Methods in Natural Language Processing (EMNLP), 2021) model (BERT-base) are used as the instance encoder. Since the instance encoder is always frozen, the embeddings of all instances are pre-computed and only keep the embeddings. In source task training, the model (skill latent space and prompt generator) is simply tuned on the mixture of all source tasks for each setting. The model is tuned for 80K steps for the high-to-low setting, and 40K steps for the other setting. In learning and testing on target tasks, the model is tuned for 100K on each target task. A checkpoint every 500 steps, which is saved and report results on the checkpoint with the highest validation performance. The prompt generator generates 64 soft tokens in all experiments. The following hyperparameters are also shared in all target and source task training: learning rate (0.3), the number of warmup steps (4000), optimizer (Adam).

For high-to-low resource transfer, Table 1 in FIG. 5 shows the example results. The proposed method (referred to as SHARPT) are compared to methods with a comparable compute- and parameter-efficiency, PROMPT-TUNING (Lester et al., the power of scale for parameter-efficient prompt tuning, in Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, pages 3045-3059, 2021) and SPOT-Retrieval (see Vu et al.). SHARPT demonstrates a clear improvement over the two methods across most tasks and on the average performance. SHARPT is also compared with much more expensive methods, SPOT-Oracle (see Vu et al.) and MODEL TUNING. Note that SPOT-Oracle is significantly more expensive than SHARPT method since it tunes on each target task with each possible task prompt (e.g., it requires roughly 48 times more training time). While being much more efficient, SHARPT matches or outperforms SPOT-Oracle. Also, SHARPT performance is on par with the MODEL TUNING

performance which requires to tune the entire model. Together, these results indicate SHARPT is an efficient and competitive approach.

In transfer across different task categories, the results are shown in Table 2 of FIG. 6. SHARPT outperforms both PROMPT TUNING and SPOT methods. The improvement over SPOT methods is larger in this setting than in the high-to-low transfer setting. This might be because SPOT relies more on knowledge shared by tasks within the same category, while SHARPT learns a shared latent space across all source tasks and is more able to leverage knowledge shared across datasets of different categories.

In across model scales, SHARPT can close the performance gap between full model tuning and prompt-based methods on a mid-sized model, T5-base (220M). Here conducts experiments with larger models, T5-large (800M) and T5-xl (3B) and compare SHARPT to MODEL TUNING and PROMPT TUNING. As shown in FIG. 8, SHARPT matches or slightly outperforms MODEL TUNING under the three model scales. Our method also shows considerable improvements over PROMPT TUNING.

Two key components of SHARPT are ablated: (1) training on source tasks; (2) skill latent space that captures shared knowledge. Example results are shown in Table 3 of FIG. 7. Clearly, knowledge learned from source tasks and encoded in the latent space is critical for target task performance.

For task relations, it is investigated if the latent space captures source and target task relations to allow knowledge transfer. Each instance queries the latent space and selects one latent skill. This selection is converted to a one-hot vector and treat it as an instance encoding. A task representation is the average of instance encodings in the task. The cosine similarity between two task representations is computed as their relation. The relations between source and target tasks are visualized in FIG. 9. It seems that more complicated source tasks such as QA and NLI tasks transfer more knowledge to target tasks via the skill latent space.

This description and the accompanying drawings that illustrate inventive aspects, embodiments, implementations, or applications should not be taken as limiting. Various mechanical, compositional, structural, electrical, and operational changes may be made without departing from the spirit and scope of this description and the claims. In some instances, well-known circuits, structures, or techniques have not been shown or described in detail in order not to obscure the embodiments of this disclosure. Like numbers in two or more figures represent the same or similar elements.

In this description, specific details are set forth describing some embodiments consistent with the present disclosure. Numerous specific details are set forth in order to provide a thorough understanding of the embodiments. It will be apparent, however, to one skilled in the art that some embodiments may be practiced without some or all of these specific details. The specific embodiments disclosed herein are meant to be illustrative but not limiting. One skilled in the art may realize other elements that, although not specifically described here, are within the scope and the spirit of this disclosure. In addition, to avoid unnecessary repetition, one or more features shown and described in association with one embodiment may be incorporated into other embodiments unless specifically described otherwise or if the one or more features would make an embodiment non-functional.

Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be

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employed without a corresponding use of other features. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. Thus, the scope of the invention should be limited only by the following claims, and it is appropriate that the claims be construed broadly and, in a manner, consistent with the scope of the embodiments disclosed herein.

What is claimed is:

1. A method of shared prompt-tuning in language models, the method comprising:
 - receiving, via a data interface, a training dataset of instances;
 - encoding, by an instance encoder, a first instance from the training dataset into a first embedding in a latent space;
 - querying a skill latent space on the first embedding to obtain a first skill vector;
 - transforming, by a prompt generator, the first skill vector into a set of prompt tokens;
 - generating, by a pretrained language model, an answer in response to an input of the first instance prepended with the prompt tokens;
 - computing a first training objective as a cross-entropy loss between a conditional distribution of the generated answer conditioned on the prompt generator and the first instance and a first label corresponding to the first instance from the training dataset;
 - computing a second training objective based on a difference between a stop gradient operator over the first embedding and the first skill vector;
 - computing a third training objective as a weighted sum of the first training objective and the second training objective; and
 - updating the prompt generator based on the third training objective via backpropagation while the pre-trained language model and the instance encoder are frozen.
2. The method of claim 1, wherein the training dataset includes instances and corresponding labels, wherein each pair of instance and corresponding label corresponds to a source natural language processing (NLP) task or a target NLP task.
3. The method of claim 1, wherein the querying the skill latent space includes:
 - retrieving a tunable set of skill vectors in the latent space; and
 - querying the tunable set of skill vectors on the first embedding by determining a nearest neighbor to the first embedding as a first skill vector for the first instance.
4. The method of claim 1, further comprising:
 - updating both the prompt generator and the tunable set of skill vectors based on the third training objective via backpropagation while the pre-trained language model and the instance encoder are frozen, when the first instance is selected from a mixture of source NLP tasks.
5. The method of claim 1, further comprising:
 - updating only the prompt generator based on the first training objective via backpropagation while the tunable set of skill vectors, the pre-trained language model and the instance encoder are frozen, when the first instance is selected only for the target NLP task.
6. The method of claim 1, wherein the prompt generator is a light-weight multilayer perceptron (MLP).
7. The method of claim 1, wherein the set of prompt tokens are generated by at least a tangent transformation of a linear transformation of the first skill vector.

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8. The method of claim 1, further comprising:

- pre-computing, by the instance encoder, embeddings for instances in the training dataset; and
- retrieving the first embedding for the first instance at training stage.

9. A system of shared prompt-tuning in language models, the system comprising:
 - a data interface that receives a training dataset of instances;
 - a memory storing an instance encoder, a skill vector generator, a prompt generator and a pretrained language model, and a plurality of processor-executable instructions; and
 - one or more processors reading and executing the plurality of processor-executable instructions to perform:
 - encoding, by the instance encoder, a first instance from the training dataset into a first embedding in a latent space;
 - querying a skill latent space on the first embedding to obtain a first skill vector;
 - transforming, by the prompt generator, the first skill vector into a set of prompt tokens;
 - generating, by the pretrained language model, an answer in response to an input of the first instance prepended with the prompt tokens;
 - computing a first training objective as a cross-entropy loss between a conditional distribution of the generated answer conditioned on the prompt generator and the first instance and a first label corresponding to the first instance from the training dataset;
 - computing a second training objective based on a difference between a stop gradient operator over the first embedding and the first skill vector;
 - computing a third training objective as a weighted sum of the first training objective and the second training objective; and
 - updating the prompt generator based on the third training objective via backpropagation while the pre-trained language model and the instance encoder are frozen.
10. The system of claim 9, wherein the training dataset includes instances and corresponding labels, wherein each pair of instance and corresponding label corresponds to a source natural language processing (NLP) task or a target NLP task.
11. The system of claim 9, wherein an operation of querying the skill latent space includes:
 - retrieving a tunable set of skill vectors in the latent space; and
 - querying the tunable set of skill vectors on the first embedding by determining a nearest neighbor to the first embedding as a first skill vector for the first instance.
12. The system of claim 9, wherein the operations further comprise:
 - updating both the prompt generator and the tunable set of skill vectors based on the third training objective via backpropagation while the pre-trained language model and the instance encoder are frozen, when the first instance is selected from a mixture of source NLP tasks.
13. The system of claim 9, wherein the operations further comprise:
 - updating only the prompt generator based on the first training objective via backpropagation while the tunable set of skill vectors, the pre-trained language model and the instance encoder are frozen,

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when the first instance is selected only for the target NLP task.

14. The system of claim 9, wherein the prompt generator is a light-weight multilayer perceptron (MLP).

15. The system of claim 9, wherein the set of prompt tokens are generated by at least a tangent transformation of a linear transformation of the first skill vector.

16. A non-transitory processor-readable storage medium storing a plurality of processor-executable instructions for shared prompt-tuning in language models, the instructions being executed by one or more processors to perform operations comprising:

receiving, via a data interface, a training dataset of instances;

encoding, by an instance encoder, a first instance from the training dataset into a first embedding in a latent space;

querying a skill latent space on the first embedding to obtain a first skill vector;

transforming, by a prompt generator, the first skill vector into a set of prompt tokens;

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generating, by a pretrained language model, an answer in response to an input of the first instance prepended with the prompt tokens;

computing a first training objective as a cross-entropy loss between a conditional distribution of the generated answer conditioned on the prompt generator and the first instance and a first label corresponding to the first instance from the training dataset;

computing a second training objective based on a difference between a stop gradient operator over the first embedding and the first skill vector;

computing a third training objective as a weighted sum of the first training objective and the second training objective; and

updating the prompt generator based on the third training objective via backpropagation while the pre-trained language model and the instance encoder are frozen.

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